

Climate Change and Avian Disease

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10.1 Introduction

Rapid and accelerating changes in global temperature, precipitation patterns, and other abiotic variables are having profound effects on biodiversity. As a diverse, highly visible, and globally distributed taxon, avian responses to ongoing global change have been relatively well documented. The capacity for long-distance flight and dietary or habitat flexibility have allowed some species, such as the blackcap (*Sylvia atricapilla*) and cattle egret (*Bubulcus ibis*), to rapidly colonize novel seasonal environments (Berthold et al. 1992; Maddock and Geering 1994). In spite of this, avian communities struggle to keep pace with geographic and phenological shifts in suitable environmental conditions (Devictor et al. 2008). Recent analyses confirm that birds are experiencing widespread declines, particularly pronounced among range-restricted habitat specialists, such as grassland birds, and long-distance migrants, such as shorebirds (Rosenberg et al. 2019). Together, these declines lend urgency to understanding the abiotic and biotic factors determining whether species will adapt or perish under ongoing and future climate change.

In addition to the direct fitness consequences of changing abiotic conditions on birds, climate change alters the arena of species interactions, including exposure to infectious pathogens. There are several high-profile instances of climate change influencing the dynamics of avian diseases of conservation concern. The parasite responsible for avian malaria, *Plasmodium relictum*, was introduced into the Hawaiian islands by human activity along with its primary mosquito vector, *Culex quinquefasciatus*, devastating immunologically naïve native

songbird communities (Warner 1968). Climate change continues to support the upslope march of *C. quinquefasciatus* by expanding larval habitat availability and temperatures favoring mosquito survival and parasite development (Atkinson et al. 2014). In late 2019, efforts to introduce critically endangered kiwī (*Pseudonestor xanthophrys*) to recently restored habitat were thwarted when at least 9 of 13 birds died from avian malaria within a few weeks of translocation (Warren et al. 2020). On Kauai, a lack of high-elevation sites suggests that it is only a matter of time before malaria reaches the last remaining disease-free refugia for several Hawaiian endemics, including the threatened 'i'iwi (Figure 10.1) (Paxton et al. 2016).

Increasingly, pathogens harbored by birds and other wildlife emerge and spill over into humans and domestic animals, motivating studies of how climate change might influence zoonotic spillover risk (see Chapter 12). One high-profile example is highly pathogenic avian influenza viruses, which can cause high human mortality and devastating economic losses to poultry and which are naturally maintained in wild waterbirds (Webster et al. 1992; see Chapter 11). Changes in environmental conditions throughout waterbird annual ranges may promote increased intercontinental mixing in Arctic breeding grounds and stopover sites and increased contact between domestic and wild birds using agricultural lands in the non-breeding season, both of which could provide novel opportunities for viral reassortment and cross-species transmission (Morin et al. 2018; Figure 10.1). Complicating predictions of future risk, warming water temperatures can reduce environmental viral persistence (Brown et al. 2009) and promote increased residency in

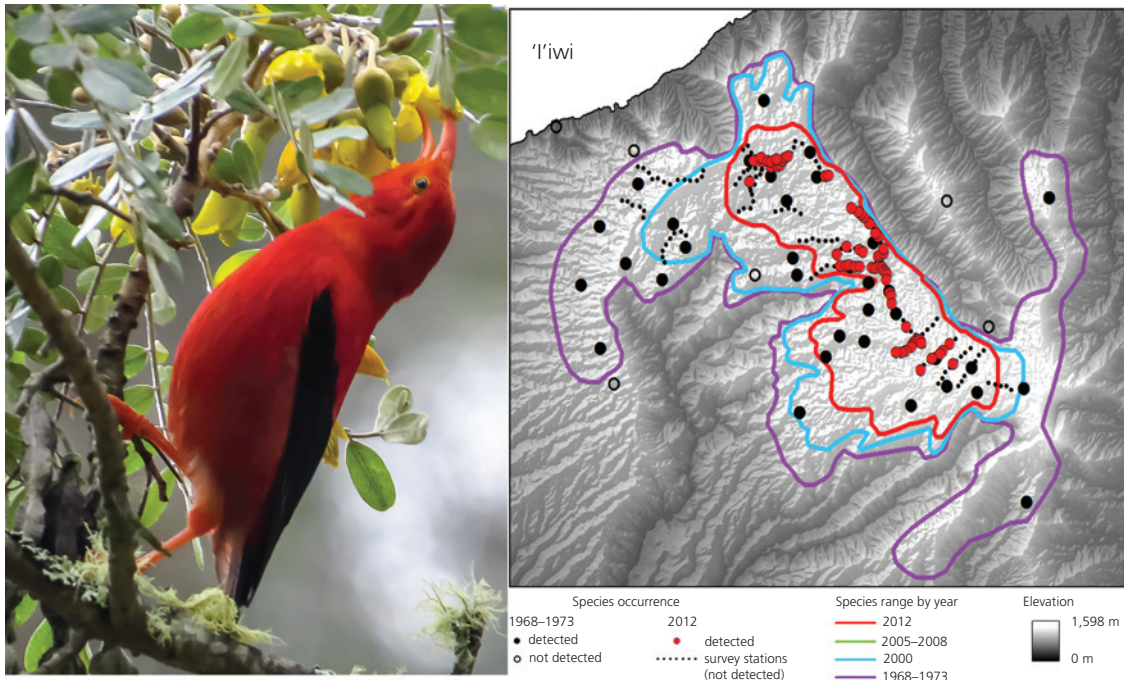


Figure 10.1 Climate warming and altered precipitation is increasing transmission of the parasite *Plasmodium relictum* that causes avian malaria in high-elevation Hawaiian forests, threatening the persistence of native Hawaiian honeycreepers, including the 'i'iwi (*Vestiaria coccinea*). Map shows estimated range contraction of 'i'iwi on Kauai based on count data from 1968 (purple), 2000 (blue), and 2012 (red). (Photo credit: R. Hall. Figure from Paxton et al. 2016.)

some waterfowl (Spivey et al. 2017). Understanding how climate change simultaneously influences host and pathogen ecology is crucial for predicting whether parasitism is likely to increase or decrease in the future.

The goal of this chapter is to survey theoretical predictions and empirical evidence for how climate change will affect avian host–parasite interactions through changes in bird distributions and timing of life history events that influence exposure and transmission (see Section 10.2); through shifts in host physiology and behavior that influence susceptibility and impacts of infection (see Section 10.3); and through effects on the replication and persistence of diverse parasite types outside of avian hosts (i.e., in vectors or the external environment; see Section 10.4). Finally, I will highlight key knowledge gaps and future research directions aimed at improving capacity to anticipate and effectively respond to shifting patterns of avian diseases (see Section 10.5).

10.2 Infection consequences of changes in bird distribution and phenology

Climate change is shifting the distribution and timing of environmental conditions that maximize breeding success and non-breeding survival for birds. Among the commonest and most readily observable responses to climate change across taxa are range shifts and/or shifts in the timing of habitat use and events in the annual cycle (Cotton 2003; La Sorte and Thompson 2007); however, there is much variation among bird species in their propensity to undergo these shifts. Whether birds colonize new environments or continue to occupy their historic seasonal ranges, most are facing altered environmental conditions and novel species assemblages, which could influence the diversity of infectious agents to which they are exposed. Relative to interactions with larger-bodied organisms, however, the effects of shifts in distribution and phenology on

host exposure to and transmission of infectious agents have been less thoroughly explored.

10.2.1 Latitudinal and elevational shifts in avian breeding ranges

Many bird species are undergoing climate-induced shifts in their breeding ranges. Increased temperatures underlie range shifts predominantly towards the poles (Hitch and Leberg 2007) or to higher elevations (Freeman et al. 2018); however, changes in precipitation patterns in some montane regions can lead to downslope range shifts (Tingley et al. 2012). Ecological theory suggests that species adapted to harsh conditions at high elevations and latitudes benefit from reduced diversity of natural enemies, including parasites (Greiner et al. 1975), and thus these range shifts can result in novel contact zones between species and the potential for acquiring novel parasites. For example, bidirectional parasite-sharing of avian malaria parasite lineages has been documented in the contact zones of range-expanding melodious warblers (*Hippolais polyglotta*) and historically breeding congeneric icterine warblers (*Hippolais icterina*) (Reullier et al. 2006).

On the other hand, colonizing species are predicted to experience lower levels of parasitism through two mechanisms. First, the enemy release hypothesis suggests that successfully colonizing species may benefit from a lack of coevolved natural enemies in their new range, which may increase their competitive ability (Keane and Crawley 2002). This hypothesis predicts that colonizers leave behind parasites from their historic breeding range, as observed in avian malaria parasite infections in introduced versus native house sparrows (*Passer domesticus*) (Marzal et al. 2011). Second, colonizing species may experience reduced parasite exposure risk, since avian communities at higher latitudes and elevations often harbor lower parasite diversity and prevalence, as seen with avian malaria in the tropical highlands (Zamoria-Vilchis et al. 2012) and at high latitudes in Alaska (Loiseau et al. 2012). However, there are cases in which colonizing species appear to experience neither enemy release nor reduced exposure. For example, range-expanding pied flycatchers (*Ficedula hypoleuca*) maintained

their diversity of specialist avian malaria parasites while also acquiring more new lineages from congeneric collared flycatchers (*Ficedula albicollis*) in their contact zones (Jones et al. 2018).

Of particular concern is the potential for range-expanding species to introduce novel parasites against which recipient host communities have no prior exposure history. Lack of adaptation to these introduced parasites could result in high virulence to endemic species and, in turn, increase the competitive advantage for colonizing species, often referred to as the novel weapons hypothesis (Prenter et al. 2004). Notable examples that were not generated by climate change *per se* include the high mortality and large-scale declines in Hawaiian honeycreepers (Drepanididae spp.) following introduction of *Plasmodium relictum* (Atkinson et al. 1995; Van Riper et al. 1986) and mortality in endemic Mauritius pink pigeons (*Nesoenas mayeri*) resulting from trichomoniasis (Bunbury et al. 2007). Historically, low parasite pressure near the poles could render polar species especially vulnerable to the virulent effects of parasites, as seen in wild Magellanic penguins (*Spheniscus magellanicus*) brought into zoos that subsequently experienced high mortality from avian malaria (Fix et al. 1988). The availability of immunologically naïve but competent hosts could also facilitate transmission of generalist parasites that pose risks to public health, including avian influenza and West Nile virus (Fuller et al. 2012). Thus, importation of generalist pathogens by colonizers into high-latitude, high-elevation refugia may exacerbate extinction risk of many cool-adapted species under warming.

By contrast, birds whose breeding ranges are contracting under climate warming may not face increased exposure risk to parasites. Extant populations in the core breeding range may remain in cool, parasite-free refugia, while populations at the ‘warmer,’ trailing range edge could decline below densities necessary for sustained transmission (May and Anderson 1979). Trends in parasitism at the trailing range edge remain understudied in avian systems; however, studies of parasite coextinction in seabird ectoparasites suggest that specialized parasites are likely to be extirpated as host populations contract and decline (Koh et al. 2004).

10.2.1.1 Changes in non-breeding range and movement patterns

Climate change could also impact avian disease dynamics by altering migration patterns and non-breeding distributions (Figure 10.2). Bird migration intimately shapes host–parasite interactions and has been proposed as a mechanism for reducing parasite transmission through the avian annual cycle. Escaping high parasite pressure by breeding in high-latitude, parasite-poor habitats has been suggested as a selective driver of migration (Piersma 1997), while seasonal departure from pathogen-contaminated habitats could reduce transmission opportunities (‘migratory escape,’ Altizer et al. 2011). Since migration can impose extreme energetic demands and physiological changes, migration can also reduce infection prevalence in hosts by reducing the survival probability of infected individuals (‘migratory culling,’ Altizer et al. 2011). Sedentary dark-eyed juncos (*Junco hyemalis*) exhibit higher prevalence of haemosporidian (blood parasite) infections than long-distance migrants that co-occur on their wintering grounds (Slowinski et al. 2018), providing indirect support for these mechanisms. Further, mathematical models demonstrate that host migration strategy is a crucial determinant of the population impacts of highly transmissible, virulent pathogens (Hall et al. 2014) and that migratory propensity may be shaped

by parasite exposure over evolutionary time (Shaw et al. 2019). Therefore, climate-induced changes in migratory propensity could have important consequences for parasite transmission and impacts.

Climate warming is reducing winter harshness in many regions; consequently, the wintering range of many birds is shifting polewards (La Sorte and Thompson 2007) and many migrations are shortening (Visser et al. 2009). Reduced energetic demands of shorter migrations could increase avian population sizes by reducing migratory mortality; mathematical models suggest that these higher host population sizes coupled with reduced efficacy of migratory culling can lead to explosive seasonal outbreaks and very high infection prevalence (Hall et al. 2016). Less harsh winters are enabling snow geese (*Chen caerulescens*) to overwinter at higher latitudes, where agricultural food subsidies are contributing to their rapid population growth (Gauthier et al. 2005); because these geese are an important carrier of the bacterium *Pasteurella multocida* that causes avian cholera (Samuel et al. 2005), their continued population expansion could increase the frequency of outbreaks in other co-occurring waterfowl.

Climate warming is also associated with increases in residency in many partially migratory bird species (Meller et al. 2016; Németh 2017). Relative to migrants, resident populations can support

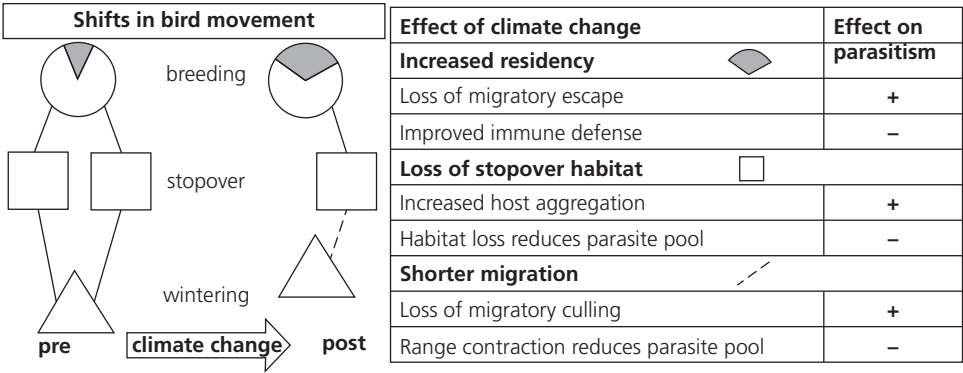


Figure 10.2 Schematic summarizing potential effects of climate change on avian host–parasite dynamics through shifts in bird movement patterns. Hypothetical migration network pre- (left) and post- (right) climate change, where a breeding site (circle) is shared by migrants and residents and lines indicate migration routes connecting to stopover habitats (squares) and an overwintering site (triangle). Climate change influences movement by increasing residency (gray shading), removing stopover habitat and reducing the distance (dashed line) to chosen overwintering site. The table lists some of the proposed mechanisms by which changing movement patterns might increase (+) or decrease (–) parasitism (as measured by infection prevalence, intensity, or diversity) under climate change.

sustained, year-round transmission and avoid migratory culling. In a model of host–parasite dynamics in partial migrants that share breeding grounds, increasing the proportion of residents elevated parasite exposure risk in returning migrants (eroding the benefits of migratory escape); because more migrants became infected, migratory culling resulted in disproportionate mortality of migrants relative to residents, suggesting that climate-induced shifts to residency could threaten the persistence of migration (Brown and Hall 2018). Mallards (*Anas platyrhynchos*) are increasingly overwintering at high latitudes and, in Alaska, maintain transmission of avian influenza virus subtypes throughout the winter (Spivey et al. 2017); thus, they may increasingly be important for the maintenance and transmission of avian influenza viruses to returning migrants (Van Hemert et al. 2014).

Changes in migration distance, routes, and habitats used for stopover and overwintering could positively or negatively influence parasite diversity in hosts. Greater migration distance and migratory propensity are positively associated with hematozoan blood parasite richness of waterfowl (Figueroa and Green 2000) and nematode diversity in shorebirds, thrushes, and hawks (Koprivnikar and Leung 2014). Climate change that results in shortened migrations could therefore reduce parasite diversity in migrants, especially if parasite-rich tropical regions are avoided (Clark et al. 2016). Different habitats used in the non-breeding period may also harbor distinct parasite communities; reduced availability or use of these habitats could therefore alter parasite richness and diversity in migrants. For example, habitat generalism is associated with increased helminth and haemosporidian richness and prevalence in migratory shorebirds relative to those using only saltwater habitats (Clark et al. 2016; Gutierrez et al. 2017). Increased drought frequency that reduces availability of inland stopover sites could reduce infection prevalence if more shorebirds switch to migrating along coastal routes with lower parasite richness. Alternatively, if loss of intertidal mudflats through sea level rise increases use of inland sites with higher parasite diversity, aggregation of migrants in remaining stopover habitats could enhance transmission risk.

Continental and global scale changes to air and ocean currents or to habitats that act as physical barriers to migration could shift migration routes and increase connections between formerly isolated flyways, with implications for the long-distance spread and introduction of novel pathogens, including zoonoses (see Chapter 12). For example, transoceanic movements of Arctic-breeding waterfowl and shorebirds transport avian influenza virus lineages between Eurasia and North America (Winker and Gibson 2010). Longer periods of ice-free water in the Arctic could therefore increase the frequency of parasite transfer and the potential for strain reassortment between flyways and continents. Overall, shifts in migration routes will alter the community of parasites to which migrants are exposed, including the potential for birds to spread pathogens into new geographic regions and new hosts.

10.2.2 Changes in timing of life history events through the annual cycle

Climate change is shifting the timing of suitable environmental conditions and resource availability; associated shifts in the timing of life history events could alter parasite transmission via seasonal changes to host exposure and susceptibility. For example, in some species, climate change is favoring advanced arrival at breeding grounds, advanced breeding, and thus longer breeding seasons, with the potential for multiple broods (Cooper et al. 2011; Smith and Moore 2005). Migrants able to advance their breeding arrival dates could face an extended window for transmission of generalist parasites harbored by resident birds and vector-borne diseases (Brown and Hall 2018; Møller 2010). Birds able to increase the number of broods, or re-nesting probability, in response to an extended breeding season could experience more parasite transmission and impacts through an increase in the number of immunologically naïve young birds. For example, in barn swallows (*Hirundo rustica*), parasitism by nest ectoparasites is thought to increase with the number of broods, which in turn influences the quality and quantity of offspring raised over a season (de Lope and Møller 1993).

Alternatively, bird populations less able to track advances in resource phenology, such as

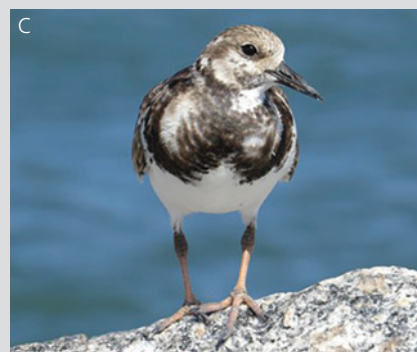
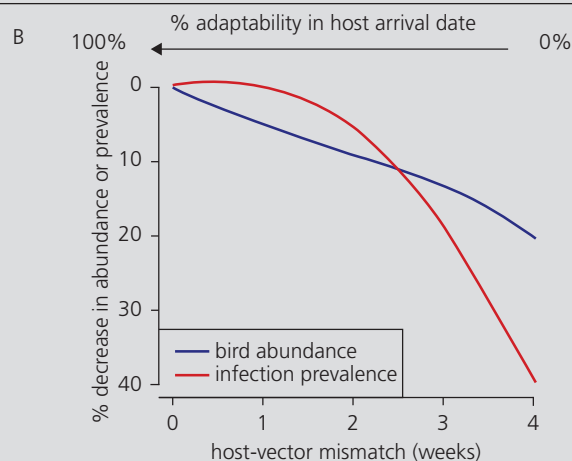
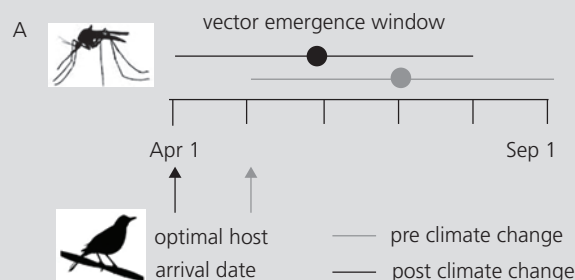
long-distance migrants (Saino et al. 2010), could experience reductions in transmission due to shifts in resource availability. In a mathematical model of vector-borne parasite transmission at a migrant's breeding ground, infection prevalence decreased with the degree of mismatch between host arrival and advancing resource phenology, because mismatch reduced host density and advancing emergence of vectors reduced their overlap with returning migrants (host–parasite mismatch; Hall et al. 2016; Box 10.1). Similarly, a model investigating the

effects of mismatch between horseshoe crab spawning and ruddy turnstone (*Arenaria interpres*) arrival on transmission of avian influenza virus in Delaware Bay predicted an increase in the probability of viral extinction with mismatch (Brown and Rohani 2012). In some systems, parasite infection can alter the timing of movement in infected hosts, delaying arrival at breeding sites (Møller et al. 2004) or increasing the duration of stopover (Hegemann et al. 2018). Climate change could exacerbate the inability of infected birds to track resource

Box 10.1 Modeling effects of climate change on bird migration and its consequences for host–parasite interactions

(A,B) In this hypothetical scenario explored by Hall et al. (2016), birds differ in their ability to track climate-mediated advances in resource phenology, potentially leading to population declines through phenological mismatch and altered overlap with reservoir hosts and disease vectors.

(A) Climate change at the migrant's breeding site shifts the peak (black circle) and duration (black horizontal line) of biting vector emergence and optimal arrival date (black vertical arrow) for returning migrants to match breeding resources by one month relative to historical phenology (gray).



(B) The extent to which migratory insectivores update their arrival date at breeding sites to track resource phenology and vector emergence (host–vector mismatch, x-axis), with ‘adaptability’ of arrival date decreasing left to right, influences the migrant’s peak population size (blue line) and peak infection prevalence (red line) of a vector-borne pathogen over an annual cycle. Changes in population size and infection prevalence are shown as percent declines relative to their respective values assuming no mismatch (i.e., host arrival perfectly tracks resource and vector phenology). Population size and infection prevalence in the migrant both decline with increasing phenological mismatch with invertebrate prey and biting vectors (e.g., decreasing adaptability); however, migrant population declines are more modest than declines in prevalence under high mismatch (low adaptability) because reduced recruitment from resource mismatch is offset by reduced disease-induced mortality. This scenario illustrates the need to consider the diverse effects of climate on fitness-relevant traits to understand the combined implications of climate change and pathogens for avian populations.

(C) In a model of avian influenza virus (AIV) transmission at a spring stopover site of ruddy turnstones (*Arenaria interpres*), Brown and Rohani (2012) found that if turnstones advanced their arrival time to match spawning horseshoe crab phenology, AIV transmission increased due to increased overlap with resident and wintering ducks; alternatively, if only horseshoe crab phenology advanced, resulting declines in ruddy turnstones caused AIV prevalence to drop. Photo credit: R. Hall.

(D) In a model of transmission of the haemosporidian *Leucocytozoon fringillarium* by blackflies (*Simulium* spp.) at high-elevation breeding sites of white-crowned sparrows (*Zonotrichia leucophrys*), Murdock et al. (2013) explored how advancing blackfly (but not sparrow) phenology influenced transmission. They found that a small advance in blackfly phenology increased parasite transmission due to increased overlap of blackflies with relapsing birds at the start of the breeding season; however, further advances in phenology reduced transmission due to reduced overlap between blackflies and parasite-susceptible fledglings. Photo credit: R. Hall

phenology, which could lower infection prevalence through increased mortality of infected individuals and increased temporal separation of susceptible and infected birds that reduces transmission. Thus, climate change has the potential to have opposing effects on disease risk via changes in the timing of life history events.

10.3 Changes in bird physiology, behavior, and response to infection

Although changes in distribution and phenology are frequently observed responses to climate change, physiological and behavioral adaptations to environmental change within a species’ current range may be just as important for species’ persistence, especially those with restricted ranges and in rapidly warming regions. However, responses such as behavioral thermoregulation often involve trade-offs with other activities such as foraging (du Plessis et al. 2012). These individual-level responses, in turn, influence parasite acquisition and impacts: physiological changes may require reallocation of resources, with consequences for resistance and tolerance to infection; behavioral changes can similarly influence exposure to and onward transmission of infectious agents (see also Chapters 3 and 4).

10.3.1 Changes in immunity and infection impacts

Immune defense in birds varies latitudinally (Martin et al. 2004), potentially reflecting differential immune costs determined by local environmental conditions and parasite exposure (Møller 2010). Climate-driven shifts in these conditions could therefore alter infection dynamics and outcomes by altering within-host processes influencing competence for infection and disease-induced mortality. For example, tree swallows (*Tachycineta bicolor*) demonstrate latitudinal variation in reproductive investment, which is thought to reflect higher investment in self-maintenance over offspring quality in parts of the breeding range where interannual survival is high (Ardia 2005; Figure 10.3A). Experimental augmentation of swallow clutch size resulted in differential changes to parental nest provisioning rates and immune defense in parents and offspring. In Alaska, where adult apparent survival is low, parents increased their provisioning rates and incurred immune costs (Figure 10.3B), while nestling condition (Figure 10.3C) and immunity (Figure 10.3D) were maintained. In Tennessee, where adult swallow survival is higher, parents did not increase provisioning rates and maintained their immune

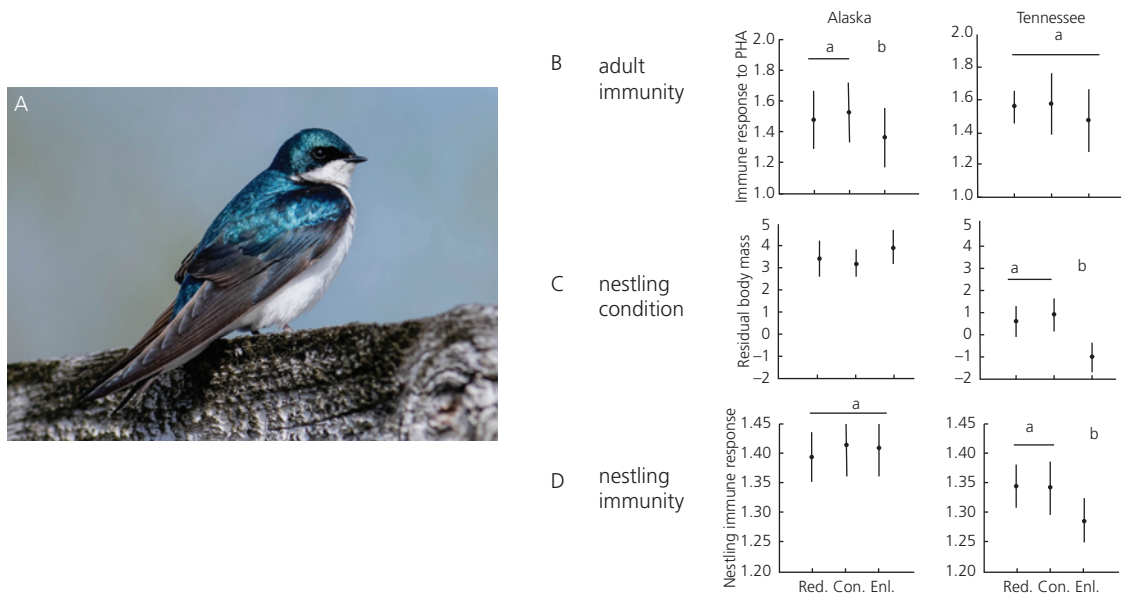


Figure 10.3 Geographic variation in trade-offs between immune and reproductive investment in tree swallows (*Tachycineta bicolor*; A) suggests that climate change could alter parasite infection outcomes via changes in age-dependent immunity (Ardia 2005). Reducing ('Red.') or Enlarging ('Enl.') brood size of tree swallow nests in Alaska (left column) or Tennessee (right column) relative to controls ('Con.') demonstrated regional differences in effects of brood augmentation on (B) adult cellular immunity, (C) offspring body condition (residual body mass), and (D) nestling immune response. Alaskan females incurred an immune cost under brood augmentation but nestlings maintained similar condition and immune function across treatments, while in Tennessee, females maintained immune function but offspring condition and immunity were reduced under brood augmentation. Photo credit: J. Owen

defense, but their offspring incurred immune costs. Thus, if patterns seen in swallows also occur in other avian species, shifts in adult survival under climate change could differentially influence infection outcomes in different parts of the range; transmission could increase in areas where females prioritize self-maintenance and thus produce immunocompromised offspring and decrease in areas where they produce few high-quality offspring.

Immune defense also varies seasonally (Hegemann et al. 2012) and, in resident birds, costs of immune defense and infection may be higher in winter due to increased metabolic demands (see Chapter 3). Reduced winter harshness could reduce these energetic costs; this could either reduce transmission through improved immune defense or increase prevalence if hosts are better able to tolerate infection. In house finches (*Haemorrhous mexicanus*), additive costs of thermoregulation and *Mycoplasma gallisepticum* infection suggest that birds may need to forage more in cold weather,

which could in turn increase their exposure to fomites on feeders (Hawley et al. 2012); milder winters might therefore reduce both parasite exposure and susceptibility. Since mounting an immune defense negatively influences body mass in overwintering house sparrows (Moreno-Rueda 2011), reduced winter harshness could alter infection dynamics by modifying the trade-off between immunity and maintenance; this could reduce transmission if birds are better able to resist infection or increase transmission if infected birds better tolerate infection. By contrast, increased frequency of extreme weather events, such as heatwaves in summer, could increase the negative impact of infections on hosts; for example, the negative effects of enteric *Salmonella* infection on broiler chicken condition were exacerbated when birds were subjected to five weeks of heat stress (temperatures of 31°C; Quinteiro-Filho et al. 2012). The combined effect of heat stress and infection in wild birds remains unknown.

Shifts towards shorter migrations and residency under climate change could confer benefits to host immune defense that reduce parasite transmission or impacts. Migratory species have larger immune defense organs than related resident species (Møller and Erritzøe 1998), probably reflecting exposure to diverse pathogens throughout the migratory range; thus, migratory populations that transition to residency may be able to mount more effective immune responses. Since physiological changes associated with active migration can be immunologically costly (Owen and Moore 2008), foregoing migration could further increase immune defense over the annual cycle. Indirect evidence for this comes from partially migratory common blackbirds (*Turdus merula*), where migrants have lower baseline immune defense than residents (Eikenaar and Hegemann 2016) and are increasingly becoming resident in urban areas (Németh 2017) and where urban populations experience lower prevalence and diversity of hematozoan blood parasite infections than forest-breeding populations (Geue and Partecke 2008). Overall, there are several ways in which climate change may alter the nature of physiological trade-offs between immune defense and other fitness components, but the difficulty in quantifying these trade-offs means that empirical evidence for climate change effects on immune defense is lacking in wild birds.

10.3.2 Changes in behavior

Changes in abiotic conditions such as temperature as well as climate effects on the amount and distribution of food resources could alter seasonal patterns of aggregation in relation to pathogen exposure risk. Since house finch flock sizes tend to peak in the coldest months and increase with latitude, and the prevalence of *M. gallisepticum* in house finch flocks increases with flock size (Altizer et al. 2004), shorter periods of cold temperatures at higher latitudes could reduce infection risk and change the timing of peak infection (Hosseini et al. 2004). Communal roosting can alter pathogen transmission and persistence in opposing ways. For example, some studies of West Nile virus suggest that corvid roosts can maintain persistence of the virus over winter (Montecino-Latorre and

Barker 2018) and that American robin (*Turdus migratorius*) roosts can increase infection prevalence in mosquitoes (Diuk-Wasser et al. 2010); however, other studies on American robins have found a negative effect of roosts on transmission through encounter dilution (Krebs et al. 2014). Communal roosting provides thermoregulatory benefits, and in species such as the sociable weaver (*Philetairus socius*), lower nocturnal temperatures are associated with larger roosting groups (Paquet et al. 2016). Thus, warmer temperatures that reduce the size or duration of non-breeding communal roosts could reduce parasite transmission potential. Conversely, loss of suitable migratory stopover or overwintering habitat is aggregating birds into remaining habitats, potentially increasing transmission risk. For example, extended droughts lead to seasonal reduction of natural wetland availability, increasing concentrations of waterfowl using human-created refuges and agricultural fields; these artificially high densities may promote large outbreaks of avian cholera (Wobeser 1992).

Climate change is causing many species to undergo shifts in foraging behavior and diet, which could influence susceptibility and exposure to parasites. In gulls, decreasing availability of preferred marine prey under ocean warming has increased the proportion of human-derived food in their diet over time (Blight et al. 2015). Foraging at landfills can increase gull exposure to *Clostridium botulinum*, the bacterium responsible for botulism (Ortiz and Smith 1994), and the switch to nutritionally poor or toxin-contaminated anthropogenic food could also result in immunocompromise and susceptibility to infection (Murray et al. 2016; see also Chapter 9). Other dietary shifts could reduce exposure to trophically transmitted, complex life cycle parasites. For example, reduced bill size in red knots associated with a phenological mismatch with resources at Arctic breeding sites has resulted in a shift from foraging on invertebrates to vegetation in the African wintering range (van Gils et al. 2016), potentially reducing exposure to trematode parasites in intermediate hosts.

Climate-related shifts in daily activity budgets could also mediate parasite encounter and susceptibility. Behavioral adjustments to warming temperatures in northern bobwhite (*Colinus virginianus*)

include reduced movement and seeking taller vegetation at the hottest times of day, (Carroll et al. 2015). Increased heat and drought periods appear to be associated with increased helminth infections and bobwhite declines (Blanchard et al. 2019); although the mechanisms underlying this likely reflect multiple climate effects on the helminths and their intermediate hosts, changes in host activity patterns and habitat associations could mediate exposure risk. Direct evidence of shifting activity patterns on parasitism in birds remains undocumented.

Changes in average or extreme temperatures have the potential to modify the effectiveness of antiparasite behaviors. For example, ‘sunning,’ where birds maximize their surface area exposed to direct sunlight, can reduce ectoparasite burdens through exposure to UV and may induce behavioral fever that kills internal pathogens (Bush and Clayton 2018; see also Chapter 4). Because sunning was more frequent in ectoparasite-infected violet-green swallows (*Tachycineta thalassina*) than those from which parasites were removed by fumigation (Blem and Blem 1993) and sunning birds can exhibit signs of heat stress such as panting, sunning may represent a costly activity in hot climates. In areas where temperatures are predicted to approach the upper thermal limits of birds, a behavioral trade-off between thermoregulation and parasite clearance may result in birds forgoing parasite removal opportunities. Overall, behavioral adaptations to climate change that alter diurnal or seasonal activity patterns and aggregation or that trade off with immune defense could increase or decrease parasite transmission risk.

10.4 Effects of climate change on parasites with external transmission stages

Theory predicts that the abundance and phenology of parasites that live obligately on or in hosts will be less affected by climate change than parasites that spend part of their life cycle off of hosts (Cizauskas et al. 2017). In support of this, the abundance of lice and mites on barn swallows has changed less with climate change than the infection prevalence of blood parasites transmitted by biting insects (Møller 2010). For avian parasites that spend part of

their life cycle outside of hosts in ectothermic vectors, intermediate hosts, or the abiotic environment, responses of parasite transmission to climate change may be more strongly regulated by sensitivity of these off-host stages to environmental temperature or relative humidity.

10.4.1 Vector-borne and trophically transmitted parasites

Arthropod vectors and intermediate hosts of avian parasites are often limited in their distribution and phenology by environmental variables. Warming in temperate regions can increase transmission by increasing the number of days of the year in which biting vectors are active. Consistent with this idea, earlier emergence of mosquitoes and blackflies is associated with increasing avian malaria prevalence in barn swallows (Møller 2010). Changes in temperature and relative humidity can also allow vectors to increase their geographic range by increasing both adult survival and larval habitat availability. This has been observed in the upslope expansion of *Plasmodium relictum*-transmitting mosquitoes in Hawaii (Atkinson et al. 2014) and may underlie global increases in transmission of West Nile virus at higher latitudes (Paz 2015). In the US, moderate drought can improve larval mosquito habitat quality and increase contact between vectors and birds visiting water and has been positively associated with West Nile virus transmission (Epstein 2001); these conditions are likely to become more frequent under climate warming projections. Another mechanism by which vectors could expand their range is through transport by migratory birds (Cohen et al. 2015). At high latitudes, historically harsh winters limit the establishment of ticks introduced by returning migrants, but milder winters could allow more ticks to successfully overwinter and transmit infectious agents such as *Borrelia burgdorferi*, the bacterium responsible for Lyme disease (Ogden et al. 2008; see also Chapter 8).

Life history traits of arthropod vectors and the parasites they harbor are also sensitive to climate change (Figure 10.4A). For example, in *Culex* mosquitoes that transmit West Nile virus and *Plasmodium relictum* in birds, reductions in the length of the gonotrophic cycle (the interval between blood

meals) with temperature could also increase the intensity of transmission by increasing the number of blood meals taken from birds (Reisen et al. 2014). Warming can also speed up parasite production, as observed in trematode parasites of birds, where infected snails produce more cercariae at warmer temperatures (Poulin 2006). A strong limiting factor on transmission can be the extrinsic incubation period of the parasite (i.e. the time from parasite ingestion by the vector to the vector being infectious), which is often determined by temperature. In *P. relictum*, this extrinsic incubation period in mosquitoes varies from 6–28 days and parasite development slows substantially below 21°C (LaPointe et al. 2010). Therefore, cool, high-elevation (Atkinson et al. 2014) or high latitude (Oakgrove et al. 2014) sites can act as parasite-free refugia because few infectious vectors survive long enough to acquire and transmit the parasite. Climate warming could reduce the extent of these parasite-free refugia by speeding up parasite development and allowing sustained transmission (Figure 10.4B).

While multiple empirically supported mechanisms predict increased transmission of vector-borne pathogens under climate change, changing environmental conditions in some regions could reduce

transmission through reductions in parasite or vector thermal performance (Figure 10.4B) or by altering vector habitat suitability. For example, increased warming in tropical and subtropical lowland regions could move key traits of vectors or the parasites they harbor past their thermal maxima, as predicted for some tropical mosquito-borne diseases of humans (Mordecai et al. 2019). Increases in the frequency of extreme weather events and climatic conditions could also reduce vector populations; for example, heavy rainfall and severe drought are associated with reduced adult and larval mosquito survival, respectively, and models suggest that this could limit avian malaria transmission in Hawai'i during dry years or immediately following extreme rainfall (Ahumada et al. 2004). Finally, warming temperatures could also modify the competitive outcome of co-infecting parasites with differing thermal optima. A study of blood parasite co-infections of Alaskan birds found a negative association between *Haemoproteus* and cold-adapted *Leucocytozoon* co-infections (Oakgrove et al. 2014), suggesting that warming could shift the competitive balance in favor of *Haemoproteus*.

Overall, the non-linear dependence of many traits of parasites and vectors on temperature mean

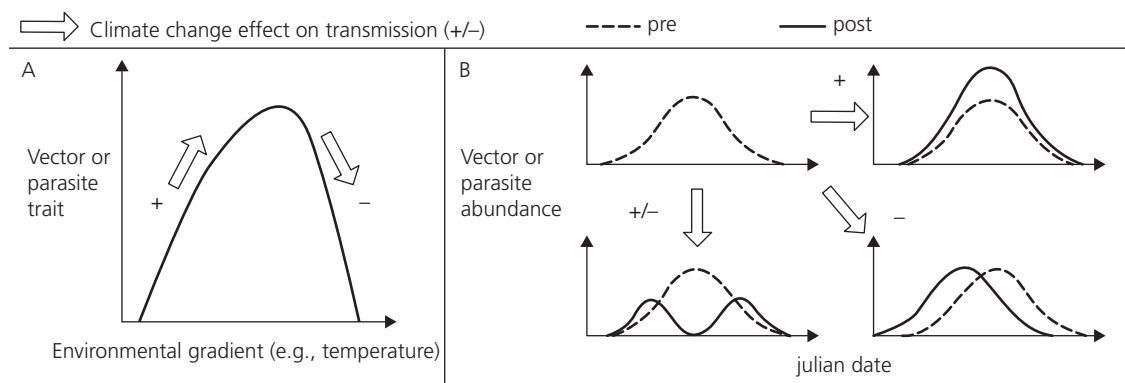


Figure 10.4 Effects of climate change (arrows) on vector or free-living parasite (A) traits and (B) abundance that could increase (+) or decrease (–) transmission in seasonal environments. (A) Many vector or parasite traits show unimodal relationships with an environmental variable, including temperature. Climate change that shifts traits towards their optima tends to increase transmission potential, while shifts away from the environmental optima will decrease transmission. (B) Seasonality in environmental conditions influences multiple traits to determine seasonal abundance of vectors or parasite propagules (top left panel, dashed line). Climate shifts towards trait optima throughout the annual cycle could increase the intensity and duration of transmission (e.g., warming in temperate regions; top right panel). Seasonal changes that move traits past their environmental tolerance (e.g., extreme summer heat in subtropical regions) could result in multiple peaks in annual transmission that increase the duration but lower the intensity of transmission (bottom left panel). Shifting phenology that advances the emergence window of vectors or parasites faster than the response in host phenology could reduce transmission via host–parasite mismatch (bottom right panel).

that moderate warming is generally expected to support increased vector-borne disease transmission in temperate, polar, and montane regions, while effects of extremes in temperature and precipitation could interrupt or prevent transmission where thermal optima are exceeded and larval habitat dries up (Figure 10.4).

10.4.2 Environmentally transmitted pathogens

For avian pathogens with externally reproducing environmental stages, a variety of environmental factors such as temperature, pH, and water depth can influence production or germination of infectious propagules. Several fungal and bacterial pathogens infect birds opportunistically and cause morbidity and mass mortality under environmental conditions that cause birds to encounter unusually high numbers of propagules. Positive associations between temperature and spore production in *Aspergillum* fungi (Seyedmousavi et al. 2015) suggest that warming might increase the number of fatal respiratory tract infections in birds. Mass mortality of waterbirds from botulism results from neurotoxins released during germination of dormant spores of *Clostridium botulinum*, and these events are associated with shallow, stagnant water and high temperatures (Rocke and Samuel 1999). Positive associations between outbreaks and water temperature in the Great Lakes (Lafrancois et al. 2011) suggest that conditions conducive to mass mortality could increase under warming.

Some pathogens have environmental stages that remain infectious for relatively short periods; therefore, changes in temperature and other environmental covariates may be crucial determinants of the seasonal buildup and duration of the transmission window. As one example, the protozoan *Trichomonas gallinae* has caused high mortality in Columbiformes and raptors where it was introduced with non-native pigeons and has emerged as a virulent pathogen of cardueline finches in Europe that co-occur at bird feeders with pigeon hosts. The persistence time of *T. gallinae* in water (up to one hour; Purple et al. 2015) has been associated with higher temperatures (Bunbury et al. 2007) and with increased humidity in bird seed (up to two days; McBurney et al. 2017); thus, regions predicted to

experience warmer and wetter climates may experience increased outbreak risk.

In contrast, some cold-adapted pathogens have persistent environmental stages lasting weeks to more than a year in favorable conditions; in these cases, warming could reduce interannual environmental persistence. For example, Newcastle disease virus shows dramatic, ten-fold declines in persistence times with increasing temperature, from 370 days at 4°C to just 6 days at 37°C (Davis-Fields et al. 2014). Similarly, the duration of infectivity in water of avian influenza viruses is associated with cool temperatures and low salinity (Brown et al. 2009). Disproportionate warming or sea level rise that increases salinity of freshwater in the Arctic breeding and staging grounds of waterfowl could therefore reduce the amount of free-living virus overwintering in Arctic waters. The resulting loss of overwintering environmental viruses could lead to transmission being largely driven by co-occurrence with shedding waterfowl, with implications for viral diversity and evolution (Hill et al. 2016).

10.5 Synthesis and future research directions

Across a diverse array of bird and parasite taxa, theoretical, field, and experimental studies of avian host-parasite interactions suggest that climate change will continue to have important and pervasive effects on parasite transmission and impacts on host populations. This review suggests that some birds, including island endemics, birds inhabiting extreme environments, and long-distance migrants, especially those that congregate in scarce and declining habitats such as wetlands, are particularly likely to face increased parasite pressure. Additionally, environmental influences on host physiology and behavior and parasite traits and transmission modes critically determine the intensity of these impacts.

The majority of empirical evidence for how climate change will influence avian host-parasite interactions examines direct or indirect effects of warming at the 'cooler' limit of host and parasite ranges, in most cases predicting increased parasitism (e.g., Garamszegi 2011). However, mathematical models and physiological theory suggest that

parasitism could decline under climate change through multiple mechanisms, including: increased parasite, vector, or infected host mortality at the 'hot' range edge (or under increasing frequency of environmental extremes); improved antiparasite defense in hosts resulting from reduced demands of surviving milder winters and shortening migrations; phenological mismatch of hosts with parasite vectors; and environmental limits to vector or parasite development or environmental pathogen persistence being exceeded (Brown et al. 2009; Eikenaar and Hegemann 2016; Murdock et al. 2013). It is unclear whether the comparative lack of evidence for climate ameliorating avian parasitism reflects true trends or publication bias towards emerging pathogens that threaten wildlife or people or geographic biases in research locations (Ducatez and Lefebvre 2014). Given that tropical regions host a high diversity of birds and parasites and historically more stable climates have led to narrower thermal niches, more empirical research documenting changes in parasitism in these regions could yield a more global perspective on climate–disease relationships.

Several studies demonstrate the potential for trade-offs between infection risk and host behavior, physiology, and other components of host fitness, which are likely mediated by the external environment. However, the logistical constraints of quantifying these trade-offs over multiple host generations mean that there is limited direct evidence for how changing climates will alter avian host competence for infection. 'Space-for-time' substitutions, where organisms are sampled from multiple locations spanning latitudinal or elevational gradients, are commonly used to provide indirect evidence for how populations will persist under future climate change (Pickett 1989), but these have less frequently been used to study avian host–parasite associations. Quantifying how host densities, immune performance, and parasite pressure relate to underlying environmental conditions in the core and extremes of current bird ranges will be crucial to identifying how these components interact to shape current and future transmission (Becker et al. 2020).

There are several limitations in relating experimental manipulations of environmental conditions to predictions for real-world avian–parasite inter-

actions under global change. First, the majority of experimental studies that quantify vector performance or environmental parasite persistence use temperature as the only covariate, while (co)variation in rainfall crucially determines bird distributions and movement and habitat suitability for vector breeding and pathogen survival (Boyle et al. 2010; LaPointe et al. 2012). Second, although vectors and parasites frequently exhibit unimodal responses to environmental covariates, experimental constraints mean that the number and range of environmental conditions at which these traits are measured are often too limited to capture this non-linearity. As a consequence, directional predictions of climate change on infection dynamics are often monotonic and region-specific and should only be cautiously extrapolated to environmental extremes and wider geographic areas. Third, extrapolating results from constant-temperature experiments to environments where temperature (and humidity) varies daily, seasonally, and by microhabitat type can underestimate effects of environmental extremes on vector or parasite survival (Carrington et al. 2013). Finally, direct effects of these environmental extremes on bird hosts in relation to infection processes are rarely quantified, with evidence particularly lacking from wild birds. Addressing these knowledge gaps will be crucial for anticipating climate-induced changes to parasite transmission and impacts.

Another major knowledge gap relates to host and parasite adaptation to climate change. The rapid pace of global climate change is probably exerting strong selection pressure on parasites, vectors, and hosts, with consequences for transmission. For example, environmental extremes that reduce expected lifespan of birds could result in selection for increased offspring production over immune defense, while reductions in seasonal harshness could promote tolerance of infection (Ardia 2005). Since parasites may be able to adapt more quickly to environmental change than their hosts, climate influences on out-of-host parasite survival or the diversity of competing parasites could lead to selection on virulence and transmissibility (Koelle et al. 2005). Theoretical and empirical studies investigating the consequences of plastic or evolutionary responses to climate change will be crucial for

predicting and managing parasite infections in imperiled avian populations.

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